Trying to understand the Galois correspondence

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Let k be a field, and let E be a finite extension of that field; in this post we will denote such a relationship by $k \hookrightarrow E$. We say that E is a Galois extension of k if the cardinality of the automorphism group $\mathrm{Aut}(E/k)$ of E fixing k is as large as it can be, namely the degree $[E:k]$ of the extension. In that case, we call $\mathrm{Aut}(E/k)$ the Galois group of E over k and denote it also by $\mathrm{Gal}(E/k)$. The fundamental theorem of Galois theory then gives a one-to-one correspondence (also known as the *Galois correspondence*) between the intermediate extensions between E and k and the subgroups of $\mathrm{Gal}(E/k)$:

> **Theorem 1 (Fundamental theorem of Galois theory)** Let E be a Galois extension of k.
>
> - (i) If $k \hookrightarrow F \hookrightarrow E$ is an intermediate field betwen k and E, then E is a Galois extension of F, and $\mathrm{Gal}(E/F)$ is a subgroup of $\mathrm{Gal}(E/k)$.
> - (ii) Conversely, if H is a subgroup of $\mathrm{Gal}(E/k)$, then there is a unique intermediate field $k \hookrightarrow F \hookrightarrow E$ such that $\mathrm{Gal}(E/F) = H$; namely F is the set of elements of E that are fixed by H.
> - (iii) If $k \hookrightarrow F_1 \hookrightarrow E$ and $k \hookrightarrow F_2 \hookrightarrow E$, then $F_1 \hookrightarrow F_2$ if and only if $\mathrm{Gal}(E/F_2)$ is a subgroup of $\mathrm{Gal}(E/F_1)$.
> - (iv) If $k \hookrightarrow F \hookrightarrow E$ is an intermediate field between k and E, then F is a Galois extension of k if and only if $\mathrm{Gal}(E/F)$ is a normal subgroup of $\mathrm{Gal}(E/k)$. In that case, $\mathrm{Gal}(F/k)$ is isomorphic to the quotient group $\mathrm{Gal}(E/k)/\mathrm{Gal}(E/F)$.

> **Example 2** Let $k = \mathbf{Q}$, and let $E = \mathbf{Q}(e^{2\pi i/n})$ be the degree $\phi(n)$ Galois extension formed by adjoining a primitive n^{th} root of unity (that is to say, E is the cyclotomic field of order n). Then $\mathrm{Gal}(E/k)$ is isomorphic to the multiplicative cyclic group $(\mathbf{Z}/n\mathbf{Z})^\times$ (the invertible elements of the ring $\mathbf{Z}/n\mathbf{Z}$). Amongst the intermediate fields, one has the cyclotomic fields of the form $F = \mathbf{Q}(e^{2\pi i/m})$ where m divides n; they are also Galois extensions, with $\mathrm{Gal}(F/k)$ isomorphic to $(\mathbf{Z}/m\mathbf{Z})^\times$ and $\mathrm{Gal}(E/F)$ isomorphic to the elements a of $(\mathbf{Z}/n\mathbf{Z})^\times$ such that $a(n/m) = (n/m)$ modulo n. (There can also be other intermediate fields, corresponding to other subgroups of $(\mathbf{Z}/n\mathbf{Z})^\times$.)

> **Example 3** Let $k = \mathbf{C}(z)$ be the field of rational functions of one indeterminate z with complex coefficients, and let $E = \mathbf{C}(w)$ be the field formed by adjoining an nth root $w = z^{1/n}$ to k, thus $k = \mathbf{C}(w^n)$. Then E is a degree n Galois extension of k with Galois group isomorphic to $\mathbf{Z}/n\mathbf{Z}$ (with an element $a \in \mathbf{Z}/n\mathbf{Z}$ corresponding to the field automorphism of k that sends w to $e^{2\pi i a/n} w$). The intermediate fields are of the form $F = \mathbf{C}(w^{n/m})$ where m divides n; they are also Galois extensions, with $\mathrm{Gal}(F/k)$ isomorphic to $\mathbf{Z}/m\mathbf{Z}$ and $\mathrm{Gal}(E/F)$ isomorphic to the multiples of m in $\mathbf{Z}/n\mathbf{Z}$.

There is an analogous Galois correspondence in the covering theory of manifolds. For simplicity we restrict attention to finite covers. If L is a connected manifold and $\pi_{L \leftarrow M} : M \to L$ is a finite covering map of L by another connected manifold M, we denote this relationship by $L \leftarrow M$. (Later on we will change our function notations slightly and write $\pi_{L \leftarrow M} : L \leftarrow M$ in place of the more traditional $\pi_{L \leftarrow M} : M \to L$, and similarly for the deck transformations $g : M \leftarrow M$ below; more on this below

the fold.) If $L \leftarrow M$, we can define $\mathrm{Aut}(M/L)$ to be the group of deck transformations: continuous maps $g : M \to M$ which preserve the fibres of π. We say that this covering map is a *Galois cover* if the cardinality of the group $\mathrm{Aut}(M/L)$ is as large as it can be. In that case we call $\mathrm{Aut}(M/L)$ the *Galois group* of M over L and denote it by $\mathrm{Gal}(M/L)$.

Suppose M is a finite cover of L. An *intermediate cover* N between M and L is a cover of N by L, such that $L \leftarrow N \leftarrow M$, in such a way that the covering maps are compatible, in the sense that $\pi_{L \leftarrow M}$ is the composition of $\pi_{L \leftarrow N}$ and $\pi_{N \leftarrow M}$. This sort of compatibilty condition will be implicitly assumed whenever we chain together multiple instances of the $\leftarrow$ notation. Two intermediate covers N, N' are *equivalent* if they cover each other, in a fashion compatible with all the other covering maps, thus $L \leftarrow N \leftarrow N' \leftarrow M$ and $L \leftarrow N' \leftarrow N \leftarrow M$. We then have the analogous Galois correspondence:

Theorem 4 (Fundamental theorem of covering spaces) Let $L \leftarrow M$ be a Galois covering.

- (i) If $L \leftarrow N \leftarrow M$ is an intermediate cover betwen L and M, then M is a Galois extension of N, and $\mathrm{Gal}(M/N)$ is a subgroup of $\mathrm{Gal}(M/L)$.
- (ii) Conversely, if H is a subgroup of $\mathrm{Gal}(M/L)$, then there is a intermediate cover $L \leftarrow N \leftarrow M$, unique up to equivalence, such that $\mathrm{Gal}(M/N) = H$.
- (iii) If $L \leftarrow N_1 \leftarrow M$ and $L \leftarrow N_2 \leftarrow M$, then $L \leftarrow N_1 \leftarrow N_2 \leftarrow M$ if and only if $\mathrm{Gal}(M/N_2)$ is a subgroup of $\mathrm{Gal}(M/N_1)$.
- (iv) If $L \leftarrow N \leftarrow M$, then N is a Galois cover of L if and only if $\mathrm{Gal}(M/N)$ is a normal subgroup of $\mathrm{Gal}(M/L)$. In that case, $\mathrm{Gal}(N/L)$ is isomorphic to the quotient group $\mathrm{Gal}(M/L)/\mathrm{Gal}(M/N)$.

Example 5 Let $L = \mathbf{C}^\times := \mathbf{C} \setminus \{0\}$, and let $M = \mathbf{C}^\times$ be the n-fold cover of L with covering map $\pi_{L \leftarrow M}(w) := w^n$. Then M is a Galois cover of L, and $\mathrm{Gal}(M/L)$ is isomorphic to the cyclic group $\mathbf{Z}/n\mathbf{Z}$. The intermediate covers are (up to equivalence) of the form $N = \mathbf{C}^\times$ with covering map $\pi_{L \leftarrow N}(u) := u^m$ where m divides n; they are also Galois covers, with $\mathrm{Gal}(N/L)$ isomorphic to $\mathbf{Z}/m\mathbf{Z}$ and $\mathrm{Gal}(M/N)$ isomorphic to the multiples of m in $\mathbf{Z}/n\mathbf{Z}$.

Given the strong similarity between the two theorems, it is natural to ask if there is some more concrete connection between Galois theory and the theory of finite covers.

In one direction, if the manifolds L, M, N have an algebraic structure (or a complex structure), then one can relate covering spaces to field extensions by considering the field of rational functions (or meromorphic functions) on the space. For instance, if $L = \mathbf{C}^\times$ and z is the coordinate on L, one can consider the field $\mathbf{C}(z)$ of rational functions on L; the n-fold cover $M = \mathbf{C}^\times$ with coordinate w from Example 5 similarly has a field $\mathbf{C}(w)$ of rational functions. The covering $\pi_{L \leftarrow M}(w) = w^n$ relates the two coordinates z, w by the relation $z = w^n$, at which point one sees that the rational functions $\mathbf{C}(w)$ on L are a degree n extension of that of $\mathbf{C}(z)$ (formed by adjoining the nth root of unity w to z). In this way we see that Example 5 is in fact closely related to Example 3.

Exercise 6 What happens if one uses meromorphic functions in place of rational functions in the above example? (To answer this question, I found it convenient to use a discrete Fourier transform associated to the multiplicative action of the nth roots of unity on M to decompose the meromorphic functions on M as a linear combination of functions invariant under this action, times a power w^j of the coordinate w for $j = 0, \ldots, n-1$.)

I was curious however about the reverse direction. Starting with some field extensions $k \hookrightarrow F \hookrightarrow E$, is it is possible to create manifold like spaces $M_k \leftarrow M_F \leftarrow M_E$ associated to these fields in such a fashion that (say) M_E behaves like a "covering space" to M_k with a group $\mathrm{Aut}(M_E/M_k)$ of deck transformations isomorphic to $\mathrm{Aut}(E/k)$, so that the Galois correspondences agree? Also, given how the notion of a path (and associated concepts such as loops, monodromy and the fundamental group) play a prominent

role in the theory of covering spaces, can spaces such as M_k or M_E also come with a notion of a path that is somehow compatible with the Galois correspondence?

The standard answer from modern algebraic geometry (as articulated for instance in this nice MathOverflow answer by Minhyong Kim) is to set M_E equal to the spectrum $\mathrm{Spec}(E)$ of the field E. As a set, the spectrum $\mathrm{Spec}(R)$ of a commutative ring R is defined as the set of prime ideals of R. Generally speaking, the map $R \mapsto \mathrm{Spec}(R)$ that maps a commutative ring to its spectrum tends to act like an inverse of the operation that maps a space X to a ring of functions on that space. For instance, if one considers the commutative ring $\mathbf{C}[z, z^{-1}]$ of regular functions on $M = \mathbf{C}^\times$, then each point z_0 in M gives rise to the prime ideal $\{f \in \mathbf{C}[z, z^{-1}] : f(z_0) = 0\}$, and one can check that these are the only such prime ideals (other than the zero ideal (0)), giving an almost one-to-one correspondence between $\mathrm{Spec}(\mathbf{C}[z, z^{-1}])$ and M. (The zero ideal corresponds instead to the generic point of M.)

Of course, the spectrum of a field such as E is just a point, as the zero ideal (0) is the only prime ideal. Naively, it would then seem that there is not enough space inside such a point to support a rich enough structure of paths to recover the Galois theory of this field. In modern algebraic geometry, one addresses this issue by considering not just the set-theoretic elements of E, but more general "base points" $p : \mathrm{Spec}(b) \to \mathrm{Spec}(E)$ that map from some other (affine) scheme $\mathrm{Spec}(b)$ to $\mathrm{Spec}(E)$ (one could also consider non-affine base points of course). One has to rework many of the fundamentals of the subject to accommodate this "relative point of view", for instance replacing the usual notion of topology with an étale topology, but once one does so one obtains a very satisfactory theory.

As an exercise, I set myself the task of trying to interpret Galois theory as an analogue of covering space theory in a more classical fashion, without explicit reference to more modern concepts such as schemes, spectra, or étale topology. After some experimentation, I found a reasonably satisfactory way to do so as follows. The space M_E that one associates with E in this classical perspective is not the single point $\mathrm{Spec}(E)$, but instead the much larger space consisting of ring homomorphisms $p : E \to b$ from E to arbitrary integral domains b; informally, M_E consists of all the "models" or "representations" of E (in the spirit of this previous blog post). (There is a technical set-theoretic issue here because the class of integral domains R is a proper class, so that M_E will also be a proper class; I will completely ignore such technicalities in this post.) We view each such homomorphism $p : E \to b$ as a single point in M_E. The analogous notion of a path from one point $p : E \to b$ to another $p' : E \to b'$ is then a homomorphism $\gamma : b \to b'$ of integral domains, such that p' is the composition of p with γ. Note that every prime ideal I in the spectrum $\mathrm{Spec}(R)$ of a commutative ring R gives rise to a point p_I in the space M_R defined here, namely the quotient map $p_I : R \to R/I$ to the ring R/I, which is an integral domain because I is prime. So one can think of $\mathrm{Spec}(R)$ as being a distinguished subset of M_R; alternatively, one can think of M_R as a sort of "penumbra" surrounding $\mathrm{Spec}(R)$. In particular, when E is a field, $\mathrm{Spec}(E) = \{(0)\}$ defines a special point p_R in M_R, namely the identity homomorphism $p_R : R \to R$.

Below the fold I would like to record this interpretation of Galois theory, by first revisiting the theory of covering spaces using paths as the basic building block, and then adapting that theory to the theory of field extensions using the spaces indicated above. This is not too far from the usual scheme-theoretic way of phrasing the connection between the two topics (basically I have replaced étale-type points $p : \mathrm{Spec}(b) \to \mathrm{Spec}(E)$ with more classical points $p : E \to b$), but I had not seen it explicitly articulated before, so I am recording it here for my own benefit and for any other readers who may be interested.

— 1. Some notation on functions —

It will be convenient to adopt notation in which many of the basic hypotheses we use take the form of an associativity-type law. To this end, we will use two non-standard notations for functions (which is implicit in the usual notation for left and right group actions), sometimes referred to as reverse Polish and Polish notation (for left and right actions respectively).

Definition 7 (Polish and reverse Polish notation) Let A, B be sets. A *function $f : A \to B$ acting on the right* is a function $f : A \to B$ in which the notation for evaluating f at an element $a \in A$ is denoted af rather than the more standard $f(a)$. If $f : A \to B, g : B \to C$ are two functions acting on the right, we write $fg : A \to C$ for the composition that is more commonly denoted $g \circ f$; this way we have the associativity-type property

$$(af)g = a(fg) \tag{1}$$

for $a \in A$.

Similarly, a *function $f : B \leftarrow A$ acting on the left* is a function $f : A \to B$ in which the notation for evaluating f at an element $a \in A$ is denoted fa rather than the more standard $f(a)$. If $f : B \leftarrow A, g : C \leftarrow B$ are two functions acting on the left, we write $gf : C \leftarrow A$ for the composition that is more commonly denoted $g \circ f$; this way we have the associativity type property

$$g(fa) = (gf)a$$

for $a \in A$. We do not define a composition between a function acting on the right and a function acting on the left.

Remark 8 Functions acting on the left largely correspond to the traditional notion of a function, despite the reversed arrow in the notation $f : B \leftarrow A$; functions acting on the right can be thought of as "antifunctions", in which the composition law is reversed.

As a general rule, in this post covering maps and deck transformations will be functions acting on the left, while paths will be functions acting on the right.

When viewing the (left or right) automorphisms of a set A as a group, we use juxtaposition $(f, g) \mapsto fg$ rather than composition $(f, g) \mapsto f \circ g$ as the group law; thus, in the case of automorphisms acting on the right, the group structure we will use is the opposite of the usual composition group structure, whereas for automorphisms acting on the left the group structure agrees with the standard composition group structure. In particular, the group of right-automorphisms of a space A is canonically identified with the opposite group of the group of left-automorphisms of a space A. (Of course, the opposite of a group G is in turn canonically identified with the group itself through the inversion operation $g \mapsto g^{-1}$, but we will refrain from using this further identification in this text as it can cause some additional confusion.)

As is customary, we will adopt the notational convention of omitting parentheses whenever there is an associativity-type law to prevent any ambiguity. For instance, if $a \in A$ and $f : A \to B$, $g : B \to C$ are functions acting on the right, we may write afg without any ambiguity thanks to (1).

— 2. Covers of manifolds —

We begin by revisiting the theory of covering spaces of manifolds. To align with the way we will be thinking about Galois theory, it will be convenient to assume the existence of a (connected) base manifold B, such that all the other manifolds L, M, N we will be considering are finite extensions of that base B. Again, for simplicity (and to make the theory more closely resemble Galois theory) we will restrict attention to finite covers, though one could extend most of this discussion to infinite covers without much difficulty.

One can think of the base manifold B as a category, in which the objects are the points $b \in B$ of the manifold, and the morphisms are the paths γ in B connecting one point $b_1 \in B$ to another $b_2 \in B$, with the composition $\gamma_1\gamma_2$ of two paths γ_1 (connecting b_1 to b_2) with γ_2 (connecting b_2 to b_3) being a path connecting b_1 with b_3 formed by concatenating the two paths together. We leave the composition $\gamma_1\gamma_2$ undefined if the terminal point of γ_1 does not agree with the starting point of γ_2. We will not distinguish a path from a reparameterisation of that path (in fact we will not mention parameterisations at all); in particular, we have the associativity law

$$(\gamma_1\gamma_2)\gamma_3 = \gamma_1(\gamma_2\gamma_3).$$

We denote the space of all paths (up to reparameterisations) by Γ. One should think of Γ as being somewhat like a group, except that the multiplication law is not always defined. (More precisely, Γ is not just a category, but is in fact a groupoid, although we will not explicitly use this fact here.)

To signify the fact that a path γ has starting point b_1 and terminal point b_2, we write

$$b_1\gamma = b_2.$$

If b_1 is not the starting point of γ, we leave $b_1\gamma$ undefined. Note this way that we obtain the associativity-type law

$$b(\gamma_1\gamma_2) = (b\gamma_1)\gamma_2$$

whenever one of the two sides is well-defined (which forces the other side to be well-defined also).

Now let $B \leftarrow M$ be a finite cover with covering map $\pi_{B\leftarrow M} : B \leftarrow M$; here all covering maps are understood to act on the left. Given a point $p \in M$ lying above a base point $b \in B$ (thus $b = \pi_{B\leftarrow M}p$), and a path $\gamma \in \Gamma$ with starting point b, we may lift γ up to a path in M that starts at p and ends at some point which we will denote by $p\gamma$. This point will lie above $b\gamma$ in B, thus we have the associativity-type law

$$(\pi_{B\leftarrow M}p)\gamma = \pi_{B\leftarrow M}(p\gamma)$$

whenever one of the two sides (and hence the other) is well-defined. It is easy to see that this defines an (right) action of Γ in the sense that one has the associativity type law

$$p(\gamma_1\gamma_2) = (p\gamma_1)\gamma_2$$

whenever $p \in M$ and $\gamma_1, \gamma_2 \in \Gamma$ are such that one of the two sides (and hence the other) is well-defined.

Note that if M is connected and $p_0 \in M$, then any other point $p \in M$ can be connected to p_0 by a path in M, which is the lift of some path in B. Thus the action of Γ is transitive in the sense that for every $p_0, p \in M$ there exists $\gamma \in \Gamma$ such that $p_0\gamma = p$.

If we have finite covers $B \leftarrow L \leftarrow M$ then the actions of Γ on L and M are compatible in the sense that one has the associativity-type law

$$(\pi_{L\leftarrow M}p)\gamma = \pi_{L\leftarrow M}(p\gamma)$$

whenever one of the two sides (and hence the other) is well-defined. A deck transformation of M (viewed as a cover of L) is a continuous map $g : M \leftarrow M$ (acting on the left) such that

$$\pi_{L\leftarrow M}g = \pi_{L\leftarrow M}. \tag{2}$$

The space $\mathrm{Aut}(M/L)$ of deck transformations is clearly a group that acts on the left on M. By working locally we see that deck transformations map lifts to lifts, thus one has the associativity-type law

$$(gp)\gamma = g(p\gamma) \tag{3}$$

whenever $g \in \mathrm{Aut}(M/L)$, $p \in M$, $\gamma \in \Gamma$ are such that one of the two sides (and hence the other) is well-defined. Conversely, one can check that any map $g : M \leftarrow M$ that obeys the two properties (2), (3) is a deck transformation (even without assuming continuity of g!). Hence one can in fact take (2), (3) to be the *definition* of a deck transformation; all the topological structure is now concealed within the path groupoid Γ and its actions on L and M.

One consequence of (3) is that the action $\mathrm{Aut}(M/L)$ of this group is free: if $g \in \mathrm{Aut}(M/L)$ is such that $gp_0 = p_0$ for some point $p_0 \in M$, then we also have $gp_0\gamma = p_0\gamma$ whenever $p_0\gamma$ makes sense, and hence by transitivity $gp = p$ for all $p \in M$, so g is the identity.

In particular, if M is a degree n cover of L (so that all fibres $\pi^{-1}_{L\leftarrow M}r$ for $r \in L$ have cardinality n), then the order of the group $\mathrm{Aut}(M/L)$ cannot exceed n, since the orbit $\mathrm{Aut}(M/L)p_0$ of any given point under this free action lies in a fibre of M above L and has the same cardinality as $\mathrm{Aut}(M/L)$. If the order equals n, we say that the cover is a *Galois cover*, use $\mathrm{Gal}(M/L)$ to denote the group $\mathrm{Aut}(M/L)$,

and conclude that the action of $\mathrm{Gal}(M/L)$ is transitive on fibres. Thus, if $p_1, p_2 \in M$ are such that $\pi_{L \leftarrow M} p_1 = \pi_{L \leftarrow M} p_2$, then there exists a unique $g \in \mathrm{Gal}(M/L)$ such that $gp_1 = p_2$.

We can now prove Theorem 4. Let $L \leftarrow M$ be a degree n cover. Let $L \leftarrow N \leftarrow M$ be an intermediate cover between N and M, and suppose that M has degree d over N. A fibre $\pi^{-1}_{L \leftarrow M} r$ of M over L has degree n, and splits into n/d fibres $\pi^{-1}_{N \leftarrow M} q$ of M over N, each of which has cardinality d (so in particular d divides n). Pick one of these fibres, and pick a point p_0 in it. By the above discussion, there are precisely d elements g of $\mathrm{Gal}(M/L)$ such that gp_0 lies in the same fibre over N as p_0:

$$\pi_{N \leftarrow M} g p_0 = \pi_{N \leftarrow M} p_0.$$

Applying paths in Γ on the right and using transitivity (and associativity), we conclude that $\pi_{N \leftarrow M} g = \pi_{N \leftarrow M}$, thus $g \in \mathrm{Aut}(M/N)$. As we have located d elements of this group, this is in fact the entire group of automorphisms, so we conclude that M is a Galois cover of N and $\mathrm{Gal}(M/N)$ is a subgroup of $\mathrm{Gal}(M/L)$. This proves part (i) of the theorem.

A similar argument shows that if $L \leftarrow N_1 \leftarrow N_2 \leftarrow M$ then $\mathrm{Gal}(M/N_2)$ is a subgroup of $\mathrm{Gal}(M/N_1)$. In particular, if N_1 and N_2 are equivalent then $\mathrm{Gal}(M/N_1) = \mathrm{Gal}(M/N_2)$. Conversely, suppose that $L \leftarrow N_1 \leftarrow M$ and $L \leftarrow N_2 \leftarrow M$ are such that $\mathrm{Gal}(M/N_2)$ is a subgroup of $\mathrm{Gal}(M/N_1)$. If $p_1, p_2 \in M$ are such that $\pi_{N_2 \leftarrow M} p_1 = \pi_{N_2 \leftarrow M} p_2$, then by the Galois nature of M over N_2 we can find $g \in \mathrm{Gal}(M/N_2)$ such that $gp_1 = p_2$. But then $g \in \mathrm{Gal}(M/N_1)$, hence $\pi_{N_1 \leftarrow M} p_1 = \pi_{N_1 \leftarrow M} p_2$. Thus we can factor $\pi_{N_1 \leftarrow M} = \pi_{N_1 \leftarrow N_2} \pi_{N_2 \leftarrow M}$ for some uniquely defined function $\pi_{N_1 \leftarrow N_2} : N_1 \leftarrow N_2$. It is not hard to verify that $\pi_{N_1 \leftarrow N_2}$ is a finite covering map that is compatible with the existing covering maps, and hence $L \leftarrow N_1 \leftarrow N_2 \leftarrow M$. This proves (iii). Reversing the roles of N_1 and N_2, we conclude in particular that if $\mathrm{Gal}(M/N_1) = \mathrm{Gal}(M/N_2)$ then N_1 and N_2 are equivalent. This gives the "unique up to equivalence" portion of (ii).

To finish the proof of (ii), let H be a subgroup of $\mathrm{Gal}(M/L)$, then we can define an equivalence relation $\sim$ on M by declaring $hp \sim p$ for every $p \in M$ and $h \in \mathrm{Gal}(M/L)$. If we set N to be the quotient space $M/\sim$ with the obvious quotient map $\pi_{N \leftarrow M} : N \leftarrow M$, then one has $\pi_{L \leftarrow M} p_1 = \pi_{L \leftarrow M} p_2$ whenever $\pi_{L \leftarrow M} p_1 = \pi_{L \leftarrow M} p_2$ (which is equivalent to $p_1 \sim p_2$). Thus we may factor

$$\pi_{L \leftarrow M} = \pi_{L \leftarrow N} \pi_{N \leftarrow M}$$

for some unique map $\pi_{L \leftarrow N} : L \leftarrow N$. One may easily verify that $\pi_{L \leftarrow N}, \pi_{N \leftarrow M}$ are both covering maps, so that we have $L \leftarrow N \leftarrow M$. To finish the proof of (Ii) we need to show that $\mathrm{Gal}(M/N) = H$. Unpacking the definitions, our goal is to show that H is the set of all $g \in \mathrm{Gal}(G/L)$ such that $gp \sim p$ for all $p \in M$. Clearly if $h \in H$ then $hp \sim p$ for all $p \in M$. Conversely, suppose that $g \in \mathrm{Gal}(M/L)$ is such that $gp \sim p$ for all $p \in M$. In particular, if one fixed a point $p_0 \in M$, then $gp_0 = hp_0$ for some $h \in H$. This implies that $gp_0\gamma = hp_0\gamma$ for all $\gamma \in \Gamma$; but the action of Γ on M is transitive, and hence $g = h$, giving the claim.

Finally, we prove (iv). By parts (i), (ii), we may assume without loss of generality that $N = M/\sim$, where $p \sim p'$ if $hp = p'$ for some $h \in \mathrm{Gal}(M/N)$.

First suppose that $\mathrm{Gal}(M/N)$ is a normal subgroup of $\mathrm{Gal}(M/L)$. Then $p \sim p'$ implies $gp \sim gp'$ whenever $p, p' \in M$ and $g \in \mathrm{Gal}(M/L)$. Thus the action of $\mathrm{Gal}(M/L)$ on M descends to that of N; since the former is transitive on fibres above L, so is the latter. Hence N is a Galois extension of L. Conversely, suppose that N is Galois. Then for any $p \in M$ and $g \in \mathrm{Gal}(M/L)$, there must exist $h \in \mathrm{Gal}(N/L)$ such that $h\pi_{N \leftarrow M} p = \pi_{N \leftarrow M} gp$. Since the right action of Γ on N is transitive, h does not actually depend on p, and is uniquely specified by g. Replacing p by $g'p$ for any $g' \in \mathrm{Aut}(M/N)$, we have

$$h\pi_{N \leftarrow M} g' p = \pi_{N \leftarrow M} g g' p,$$

but $\pi_{N \leftarrow M} g' = \pi_{N \leftarrow M}$, hence

$$\pi_{N \leftarrow M} gp = \pi_{N \leftarrow M} g g' p,$$

or equivalently $gp \sim gg'p$. This implies that $gg'g^{-1} \in \mathrm{Gal}(M/N)$, thus $\mathrm{Gal}(M/N)$ is preserved by conjugation by arbitrary elements g of $\mathrm{Gal}(M/L)$. In other words, $\mathrm{Gal}(M/N)$ is a normal subgroup. The map $g \mapsto h$ described here can easily be verified to be a homomorphism from $\mathrm{Gal}(M/L)$ to $\mathrm{Gal}(M/N)$ with kernel $\mathrm{Gal}(N/L)$, so the claim (iv) follows from the first isomorphism theorem.

> **Remark 9** Let $B \leftarrow M$ be a Galois cover, let b be a point in the base B, and let p_0 be a point in the fibre $\pi_{B\leftarrow M}^{-1} b$. For any loop $\gamma \in \Gamma$ starting and ending at b (so that $b\gamma = b$, the point $p_0\gamma$ lies in the same fibre $\pi_{B\leftarrow M}^{-1} b$ of M as p_0, hence by the Galois nature of the cover there is a unique $g \in \mathrm{Gal}(M/B)$ such that
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> $$gp_0 = p_0\gamma.$$
>
> It is easy to see that homotopic deformations of γ (preserving the initial and final point b) do not affect the value of g. Thus, we have constructed a map $[\gamma] \mapsto g_{[\gamma]}$ from the fundamental group $\pi_1(B,b)$ of B at b to the Galois group $\mathrm{Gal}(M/B)$, such that
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> $$g_{[\gamma]}p_0 = p_0\gamma. \tag{4}$$
>
> As M is connected, we see that this map is surjective. From (4) and the fact that that $\mathrm{Gal}(M/B)$ acts freely on M) we also see that the map is a homomorphism. (In many texts, the notation is set up so that this correspondence is an antihomomorphism rather than a homomorphism; we have avoided this by making the paths act on the right and the deck transformations act on the left, thus implicitly introducing an order reversal when identifying the two.) Thus we see that the Galois group $\mathrm{Gal}(M/B)$ is isomorphic to a quotient of the fundamental group $\pi_1(B,b)$. However, this isomorphism is not canonical, even if one fixes the base point b, because it depends on p_0! If one replaces p_0 by another point $p_0' = hp_0$ in the same fibre for some $h \in \mathrm{Gal}(M/B)$, then the associated map $[\gamma] \mapsto g'_{[\gamma]}$, defined by
>
> $$g'_{[\gamma]}p'_0 = p'_0\gamma,$$
>
> has to be conjugate to $g_{[\gamma]}$ in order to remain compatible with (4):
>
> $$g'_{[\gamma]} = hg_{[\gamma]}h^{-1}.$$
>
> Thus, if one does not specify the reference point p_0 in the covering space M, the identification of $\mathrm{Gal}(M/B)$ with a quotient of $\pi_1(B,b)$ is only unique up to conjugation. Thus the relationship between fundamental groups and Galois groups are a little subtle, requiring one to be aware of what base points have been selected, unless one is willing to just work up to conjugacy. (See also the above-mentioned post of Kim for some further discussion of this point.)

— 3. Extensions of fields —

We can now give an analogous treatment of the Galois correspondence for finite extensions of fields.

In this setting, we will take the base space B to be the category of integral domains (with morphisms now interpreted as "paths"). Thus, points b in this space are integral domains, and the paths in this space are ring homomorphisms $\gamma : b_1 \to b_2$ from one integral domain to another, which we will think of as functions acting on the right, in particular the composition of $\gamma_1 : b_1 \to b_2$ and $\gamma_2 : b_2 \to b_3$ will be denoted $\gamma_1\gamma_2$ (rather than the more traditional $\gamma_2 \circ \gamma_1$). As before, we use the notation $b_1\gamma = b_2$ to denote the assertion that γ is a ring homomorphism from b_1 to b_2, so in particular we have the associativity type law

$$(b\gamma_1)\gamma_2 = b(\gamma_1\gamma_2)$$

whenever either side is well-defined. As previously mentioned, we will ignore all set-theoretic issues caused by the fact that B is a proper class rather than a set, and similarly for the covers of B introduced below. We let Γ denote the collection of all ring homomorphisms $\gamma : b_1 \to b_2$ connecting points in B. This is analogous to the space of collections of paths in the base manifold B in the covering space

context. Oen caveat however is that while paths in manifolds always have inverses, the same is not true in B, because not every ring homomorphism between integral domains is invertible. This complicates the situation somewhat compared to the covering space case, but it turns out that the additional level of complication is manageable.

Given a field k, we define an associated space M_k, whose points p are ring homomorphisms $p : k \to b$ to some ring $b \in B$, where we view these homomorphism as acting on the right. There is an obvious map $\pi_{B \leftarrow M_k} : B \leftarrow M_k$ (viewed as a function acting on the left) that maps a ring homomorphism $p : k \to b$ to its codomain b. If $p \in M_k, \gamma \in \Gamma, b_1, b_2 \in B$ are such that $\pi_{B \leftarrow M_k} p = b_1$ and $b_1 \gamma = b_2$, then the composition $p\gamma$ is a ring homomorphism from k to b_2 and is thus also an element of M_k. Thus Γ acts on the right on M_k in the sense that

$$(p\gamma_1)\gamma_2 = p(\gamma_1\gamma_2)$$

whenever either side is well-defined, and the action is compatible with the base B in the sense that

$$\pi_{B \leftarrow M_k}(p\gamma) = (\pi_{B \leftarrow M_k} p)\gamma$$

whenever either side is well-defined.

> **Remark 10** One can define M_R for any commutative ring R in the same fashion. If one takes R to be the integers $\mathbf{Z}$, then $M_{\mathbf{Z}}$ is in one-to-one correspondence with B, since for every integral domain b there is a unique ring homomorphism from $\mathbf{Z}$ to b. Thus we are really thinking of all our spaces M_k as being covers of $M_{\mathbf{Z}}$. In the usual scheme-theoretic language, one views the schemes $\mathrm{Spec}(k)$ as lying above the base scheme $\mathrm{Spec}(\mathbf{Z})$.

> **Remark 11** The characteristic of the field k naturally restricts the base points b that M_k actually lies above. For instance, if k has characteristic zero, then points $p : k \to b$ of M_k only lie above those base points $b \in B$ for which the unit 1_b of b has infinite order; similarly, if k has a positive characteristic q, then points of M_k only lie above base points $b \in B$ for which the unit of b has order q. This helps explain why fields of different characteristic seem to inhabit "different worlds" – they lie above disjoint (and disconnected) portions of the base space B! However, we will not exploit the field characteristic in this post.

As mentioned in the introduction, each space M_k has a distinguished point p_k, which is the identity map on k and is the representative in this formalism of $\mathrm{Spec}(k)$. This point makes M_k is "directionally connected" in the following sense: every other point $p : k \to b$ in M_k is connected to p by a path $\gamma \in \Gamma$ in the sense that $p = p_k\gamma$. Indeed, one just takes $\gamma : k \to b$ to be precisely the same homomorphism as p. (More generally, in every commutative ring R, every point $p : R \to b$ of M_R will be connected via a path from a point in $\mathrm{Spec}(R)$, namely the kernel of p which is necessarily a prime ideal since b is an integral domain. I like to think of M_R as a sort of "penumbra" lurking around the much smaller set $\mathrm{Spec}(R)$.) However, the connectedness does not flow the other way: not every point $p : k \to b$ in M_k is connected back to p_k by a path, because paths need not be invertible. (For instance, if p is an embedding of k into a larger field E, there is no way to map E back into k by a ring homomorphism while preserving k, as homomorphisms of fields have to be injective.)

Suppose we have a finite extension $k \hookrightarrow E$ of fields, then the inclusion map $\pi_{k \hookrightarrow E} : k \to E$, viewed as a function acting on the right, is a ring homomorphism. As such, it also can be viewed as a map from M_E to M_k: any point $p : E \to b$ of M_E, when composed with $\pi_{k \hookrightarrow E}$, gives rise to a point $\pi_{k \hookrightarrow E} p : k \to b$ of M_k. We relabel this composition map as $\pi_{M_k \leftarrow M_E} : M_k \leftarrow M_E$ (now viewed as a function acting on the left), thus

$$\pi_{M_k \leftarrow M_E} p = \pi_{k \hookrightarrow E} p$$

for all $p \in M_K$. By construction we see that

$$\pi_{B \leftarrow M_k} \pi_{M_k \leftarrow M_E} = \pi_{B \leftarrow M_E}$$

and similarly for any nested sequence $k \hookrightarrow F \hookrightarrow E$ of fields one has

$$\pi_{M_k \leftarrow M_F} \pi_{M_F \leftarrow M_E} = \pi_{M_k \leftarrow M_E}.$$

Now let us look at the fibres of M_E above M_k. We need the following basic result, which relates the degree of the field extension $k \hookrightarrow E$ to the degree of the cover $M_E \hookrightarrow M_k$:

> **Proposition 12** Let E be a degree n extension of a field k. Then every fibre $\pi_{k \hookrightarrow E}^{-1}(q)$ in M_E of a point q in M_k has cardinality at most n.

Proof: Unpacking all the definitions, we are asking to show that every ring homomorphism $q : k \to b$ into an integral domain b has at most n extensions to a ring homomorphism $p : E \to b$.

We induct on n. If $n = 1$ the claim is trivial, so suppose $n > 1$ and the claim has already been proven for smaller values of n. Then E is larger than k, so we may find an element α that lies in E but not k. Let m be the degree of α, thus $m > 1$, $k(\alpha)$ is a degree m extension of k and E is a degree n/m extension of $k(\alpha)$ (in particular, m divides n). By induction hypothesis, every ring homomorphism on $k(\alpha)$ has at most n/m extensions to E, so it suffices to show that $q : k \to b$ has at most m extensions to $k(\alpha)$.

Let P be the minimal polynomial of α with coefficients in k, thus P is monic with degree m, and $P(\alpha) = 0$. Any extension $\tilde{q} : k(\alpha) \to b$ of q to $k(\alpha)$ must obey the law $P_q(\tilde{q}(\alpha)) = 0$, where P_q is the monic polynomial of degree m with coefficients in b formed by applying q to each coefficient of P. As q is an integral domain, P_q has at most m roots. Thus there are at most m possible values of $\tilde{q}(\alpha)$; since $\tilde{q}$ is completely determined by q and $\tilde{q}(\alpha)$, we obtain the claim. □

> **Example 13** Let $k = \mathbf{C}(z)$ and $E = \mathbf{C}(w)$ be the fields from Example 3. The inclusion map $q : k \to E$ is a point in M_k. The fibre $\pi_{k \hookrightarrow E}^{-1}(q)$ consists of n points $p_0, ..., p_{n-1}$, with $p_a : E \to E$ being the field automorphism of E that sends w to $e^{2\pi i a/w}$ (in particular, p_0 is the distinguished point p_E of E). In contrast, the distinguished point $p_k : k \to k$ of M_k has empty fibre $\pi_{k \hookrightarrow E}^{-1}(p_k)$, since there is no ring homomorphism from E to k that fixes k.

> **Remark 14** If two points q_1, q_2 in M_k are connected by an invertible path γ, thus $q_1\gamma = q_2$ and $q_2\gamma^{-1} = q_1$, then it is easy to see that γ induces a one-to-one correspondence between the fibre $\pi_{k \hookrightarrow E}^{-1}(q_1)$ and the fibre $\pi_{k \hookrightarrow E}^{-1}(q_2)$; in particular, the two have the same cardinality. However, in contrast to the situation with covering spaces, not all paths are invertible, and so the cardinality of a fibre $\pi_{k \hookrightarrow E}^{-1}(q)$ can vary with the base q, as can already be seen in the preceding example. However the situation is better for Galois extensions, as we shall shortly see, in which all non-empty fibres are isomorphic.

We continue analysing a field extension $k \hookrightarrow E$, viewing M_E as analogous to a covering space for M_k. By analogy with covering spaces, we could define a "deck transformation" of M_E over M_k to be a map $g : M_k \to M_k$ (acting on the left) such that

$$\pi_{M_k \leftarrow M_E} g = \pi_{M_k \leftarrow M_E} \tag{5}$$

and such that

$$(gp)\gamma = g(p\gamma) \tag{6}$$

whenever $p \in M_E$, $\gamma \in \Gamma$ are such that one side (and hence the other) of the above equation is well-defined. Let us see what a deck transformation actually is. If we apply g to the distinguished point $p_E : E \to E$ of M_E, we obtain a ring homomorphism $gp_E : E \to E$. From (5) we have

$$\pi_{M_k \leftarrow M_E} gp_E = \pi_{M_k \leftarrow M_E} p_E$$

which on chasing definitions means that gp_E is the identity on k. Thus $gp_E : E \to E$ is a k-linear map; it also preserves 1, so it has trivial kernel and is thus injective. As E is finite dimensional over k, it is also surjective; hence gp_E is in fact a field automorphism of E that fixes k. The action of g on any other point

$p : E \to b$ of M_E can then be computed by writing $p = p_E\gamma$ where $\gamma = p$ is the path connecting p_E to γ and using (6) to conclude that $gp = (gp_E)\gamma$. Conversely, given any field automorphism $\phi : E \to E$ that fixes k, one can generate a deck transformation g by setting $gp = \phi p$ for all ring homomorphisms $p : E \to b$; by chasing definitions one verifies that this is a deck transformation. Thus we have a one-to-one correspondence between deck transformations and field automorphisms of E fixing k. (This correspondence feels reminiscent of the Yoneda lemma, though I was not able to find a direct link between the two.)

> **Exercise 15** Show that the correspondence between deck transformations and field automorphisms given above is a group isomorphism. (When functions are expressed in more traditional language, this correspondence is an antiisomorphism rather than an isomorphism. We are able to avoid this reversal of order by making deck transformations act on the left and field automorphisms act on the right.)

One consequence of the above correspondence is that the space $\mathrm{Aut}(M_E/M_k)$ of deck transformations of M_E over M_k is in one-to-one correspondence with the group $\mathrm{Aut}(E/k)$ of field automorphisms of E fixing k. If $k \hookrightarrow F \hookrightarrow E$ is an intermediate field, then certainly any deck transformation of M_E over M_F is also a deck transformation of M_E over M_k, so $\mathrm{Aut}(M_E/M_F)$ is a subgroup of $\mathrm{Aut}(M_E/M_k)$.

From the above discussion, we see that a deck transformation of M_E over M_k is completely determined by its action on the distinguished point p_E. In particular, the action of $\mathrm{Aut}(M_E/M_k)$ is free on the distinguished point p_E of M_E in the sense that the points gp_E for $g \in \mathrm{Aut}(M_E/M_k)$ are all distinct. As they also lie in the same fibre of p_E above M_k, we conclude from Proposition 12 that the order of $\mathrm{Aut}(M_E/M_k)$ cannot exceed the degree of the extension:

$$|\mathrm{Aut}(M_E/M_k)| \leq [E : k]. \tag{7}$$

As stated in the introduction, we call E a Galois extension of k if we in fact have equality

$$|\mathrm{Aut}(M_E/M_k)| = [E : k],$$

and then we rename $\mathrm{Aut}(M_E/M_k)$ as $\mathrm{Gal}(M_E/M_k)$. This is canonically equivalent to the usual Galois group $\mathrm{Gal}(E/k)$ over fields. Thus, if E is a degree n Galois extension of k, then implies that the fibre containing p_E above M_k must have cardinality exactly n, with $\mathrm{Gal}(M_E/M_k)$ acting freely and transitively on this fibre. In fact the same is true for every non-empty fibre:

> **Lemma 16** Let E be a degree n Galois extension of k, and let $p : E \to b$ be a point in M_E. Then the fibre $\pi^{-1}_{M_k \leftarrow M_E}(\pi_{M_k \leftarrow M_E} p)$ containing p above M_k has cardinality exactly n, and $\mathrm{Gal}(M_E/M_k)$ acts freely and transitively on this fibre. In other words, whenever $p, p' \in M_E$ are such that $\pi_{M_k \leftarrow M_E} p = \pi_{M_k \leftarrow M_E} p'$, then there is a unique $g \in \mathrm{Gal}(M_E/M_k)$ such that $gp = p'$.

Proof: As E is a field and b is an integral domain, the ring homomorphism $p : E \to b$ must be injective (it maps invertible elements to invertible elements, and hence maps non-zero elements to non-zero elements). Each Galois group element $g \in \mathrm{Gal}(M_E/M_k)$ corresponds to a different field automorphism of E, hence the n homomorphisms $gp : E \to b$, $g \in \mathrm{Gal}(M_E/M_k)$ are all distinct. This produces n distinct elements of the fibre of p; by Proposition 12 this is the entire fibre. The claim follows. □

Thanks to this lemma, we see that M_E behaves like a degree n covering map on some subset of M_k (the space of ring homomorphisms $p : k \to b$ for which b is "large enough" that there is at least one extension of p to E).

We can now prove part (i) of Theorem 1, in analogy with the covering space theory argument:

> **Proposition 17** If E is a Galois extension of k, and $k \hookleftarrow F \hookleftarrow E$ is an intermediate field, then E is a Galois extension of F.

Note that we have already demonstrated that $\mathrm{Aut}(E/F)$ is a subgroup of $\mathrm{Aut}(E/k)$, so this proposition completes the proof of Theorem 1(i).

Proof: Let n denote the degree of E over k, and m the degree of F over k, thus E is a degree n/m extension of F. Consider a non-empty fibre of M_E above M_k. By Lemma 16, this fibre consists of n points. It also can be decomposed into fibres of M_E above M_F, indexed by a fibre of M_F above M_k. By Lemma 16, the latter fibre consists of m points, while by Proposition 12, the former fibres all have cardinality at most n/m. Hence all the former fibres must have cardinality exactly n/m. In particular, the fibre of p_E above M_F has cardinality n/m. This fibre consists of those ring homomorphisms $p : E \to E$ that fix F. As discussed previously, these ring homomorphisms must be field automorphisms, and thus generate elements of $\mathrm{Aut}(E/F)$ (or $\mathrm{Aut}(M_E/M_F)$. Thus $\mathrm{Aut}(M_E/M_F)$ has cardinality at least n/m, and hence by (7) it has cardinality exactly n/m. Thus E is a Galois extension of F as required. □

Corollary 18 If $k \hookrightarrow F \hookrightarrow E$, then $F = E^{\mathrm{Gal}(E/F)}$.

Proof: By definition, every element of F is fixed by $\mathrm{Gal}(E/F)$, thus $F \subset E^{\mathrm{Gal}(E/F)}$ and hence

$$[E : F] \geq [E : E^{\mathrm{Gal}(E/F)}].$$

On the other hand, by Proposition 17 we have $[E : F] = |\mathrm{Gal}(E/F)|$. Finally, from (7) we have

$$|\mathrm{Gal}(E/F)| \leq [E : E^{\mathrm{Gal}(E/F)}].$$

Therefore all inequalities must be equalities, and $F = E^{\mathrm{Gal}(E/F)}$ as claimed. □

Now we prove Theorem 1(ii). Let H be a subgroup of $\mathrm{Gal}(M_E/M_k)$, which we can also identify with a subgroup of $\mathrm{Gal}(E/k)$. Let E^H denote the set of elements of E that are fixed by H; this is clearly an intermediate field between k and E:

$$k \hookrightarrow E^H \hookrightarrow E.$$

On the one hand, every element of H can be identified with an element of $\mathrm{Gal}(E/E^H) \equiv \mathrm{Gal}(M_E/M_{E^H})$. By Proposition 17, $\mathrm{Gal}(M_E/M_{E^H})$ has order exactly equal to $[E : E^H]$. Thus

$$|H| \leq |\mathrm{Gal}(M_E/M_{E^H})| = [E : E^H].$$

To prove the reverse inequality, we use an argument of Artin. We first need a simple lemma:

Lemma 19 (Producing an invariant vector) Let the notation be as above. Let E^m be the m-fold Cartesian product of E (viewed as a vector space over E), and let H act componentwise on E^m:

$$h(x_1, ..., x_m) := (hx_1, ..., hx_m).$$

Let V be an H-invariant subspace of E^m (i.e., $V \subset E^m$ is a vector space over E such that $hv \in V$ whenever $h \in H$ and $v \in V$). If V contains a non-zero element of E^m, then it also contains a non-zero element of $(E^H)^m$.

Proof: Let $v = (x_1, ..., x_n)$ be a non-zero element of V with a minimal number of non-zero entries. By relabeling, we may assume that x_1 is non-zero; by dividing by x_1, we may normalise $x_1 = 1$. If all the other coefficients lie in E^H then we are done; by relabeling we may thus assume without loss of generality that $x_2 \notin E^H$. By definition of E^H, we may thus find $h \in H$ such that $hx_2 \neq x_2$. The vector $hv - v$ is then a non-zero element of V with fewer non-zero entries than v, contradicting the minimality of v, and the claim follows. □

Remark 20 This argument is reminiscent of the abstract ergodic theorem from this previous blog post, except that now we minimise the ℓ^0 norm rather than the ℓ^2 norm, since the latter is unavailable over arbitrary fields E. While this lemma is simple, it seems to be the one aspect of the Galois correspondence for field extensions that does not seem to have an obvious analogue in the covering space setting; on the other hand, I do not know how to prove the fundamental theorem of

> Galois theory without this lemma or something like it (though presumably the original arguments of Galois proceed differently).

> **Corollary 21** We have $[E : E^H] \leq |H|$.

Proof: Suppose for contradiction that $[E : E^H] = m > |H|$, thus we may find elements $z_1, ..., z_m$ of E that are linearly independent over E^H. We form a vector $z = (z_1, ..., z_m) \in E^m$ out of these elements, and consider the orbit $\{hz : h \in H\}$ of this vector. Let $V = \{v \in E^m : v \cdot hz = 0 \text{ for all } h \in H\}$ be the space of vectors orthogonal to this orbit, where we use the usual dot product

$$(v_1, ..., v_m) \cdot (z_1, ..., z_m) := v_1 z_1 + ... + v_m z_m$$

in E^m. This space has codimension at most $|H|$ in E^m and thus has a non-zero vector since $m > |H|$. Clearly V is H-invariant. By Lemma 19, V contains a non-zero vector $v = (x_1, ..., x_m)$ whose entries lie in E^H. By construction, v is orthogonal to z, thus there is a non-trivial linear relation amongst the $z_1, ..., z_m$ with coefficients in E^H, contradicting linear independence. □

We thus have $H = \mathrm{Gal}(M_E/M_{E^H}) \equiv \mathrm{Gal}(E/E^H)$, giving the existence part of Theorem 1(ii); the uniqueness part is immediate from Corollary 18.

Theorem 1(iii) is also immediate from Corollary 18, so we turn to Theorem 1(iv). Let $k \hookrightarrow F \hookrightarrow E$. First suppose that $\mathrm{Gal}(E/F)$ is a normal subgroup of $\mathrm{Gal}(E/k)$. If $g' \in \mathrm{Gal}(E/F)$ and $g \in \mathrm{Gal}(E/k)$, then $gg'g^{-1} \in \mathrm{Gal}(E/F)$ fixes F, thus g' fixes gF. Thus $gF \subset E^{\mathrm{Gal}(E/F)}$, hence by Corollary 17 we have $gF \subset F$. Thus the action of $\mathrm{Gal}(E/k)$ descends to that of $\mathrm{Gal}(F/k)$; equivalently, every deck transformation of M_E above M_k descends to a deck transformation of M_F above M_k. Since the fibre of p_E above M_k has cardinality $[E : k]$, and splits into fibres of cardinality $[E : F]$ above M_F indexed by a fibre of M_F above M_k that has cardinality at most $[F : k]$ by Proposition 12, and $[E : k] = [E : F][F : k]$, we conclude that the latter fibre must have cardinality exactly $[F : k]$. As $\mathrm{Gal}(M_E/M_k)$ acts transitively on the fibre of M_k, it must also act transitively on this fibre of M_F. Thus there at least $[F : k]$ distinct deck transformations of M_F above M_k, and so F is a Galois extension of k thanks to (7), and every deck transformation of M_F above M_k arises from a deck transformation of M_E above M_k. Thus we have constructed a surjective map from $\mathrm{Gal}(M_E/M_k)$ to $\mathrm{Gal}(M_F/M_k)$ which can easily be verified to be a homomorphism. The kernel of this homomorphism consists of all deck transformations of M_E above M_k that preserve the fibres of M_E above M_F; applying the deck transformation to the distinguished point p_E, one sees in particular that the field automorphism of E associated to this transformation preserves F, and hence this transformation lies in $\mathrm{Gal}(M_E/M_F)$. Conversely every deck transformation in $\mathrm{Gal}(M_E/M_F)$ descends to the identity transformation on M_F. Thus the kernel of the homomorphism is precisely $\mathrm{Gal}(M_E/M_F)$, and the isomorphism of $\mathrm{Gal}(F/k)$ with $\mathrm{Gal}(E/k)/\mathrm{Gal}(E/F)$ then follows from the first isomorphism theorem.

Conversely, suppose that F is a Galois extension of k. Let $g' \in \mathrm{Gal}(M_E/M_F)$ and $g \in \mathrm{Gal}(M_E/M_k)$. The points $\pi_{M_F \leftarrow M_E} p_E$, $\pi_{M_F \leftarrow M_E} g p_E$ of F lie above the same point of M_k; from Lemma 16 there thus exists $h \in \mathrm{Gal}(M_F/M_k)$ such that

$$h\pi_{M_F \leftarrow M_E} p_E = \pi_{M_F \leftarrow M_E} g p_E.$$

On the other hand, as p_E is the distinguished point of M_E, there exists $\gamma \in \Gamma$ such that $p_E \gamma = g' p_E$. Applying γ on the right to the above equation, we thus have

$$h\pi_{M_F \leftarrow M_E} g' p_E = \pi_{M_F \leftarrow M_E} g g' p_E.$$

But $p_E, g' p_E$ lie above the same point of M_F, thus $\pi_{M_F \leftarrow M_E} p_E = \pi_{M_F \leftarrow M_E} g' p_E$, and hence

$$\pi_{M_F \leftarrow M_E} g g' p_E = \pi_{M_F \leftarrow M_E} g p_E.$$

By Lemma 16 again, this implies that $gg'p_E = g''gp_E$ for some $g'' \in \mathrm{Gal}(M_E/M_F)$. As p_E connects to every other point in M_E by a path in Γ, we conclude that $gg' = g''g$, thus $gg'g^{-1} \in \mathrm{Gal}(M_E/M_F)$. In other words, $\mathrm{Gal}(M_E/M_F)$ is a normal subgroup of $\mathrm{Gal}(M_E/M_k)$.

Comments (18)

Fausto di Biase · 28 August 2018

This seems reminiscent of some recent work of Peter Scholze.

> **test** · 28 August 2018
>
> No it doesn't.

Ben Spitz · 29 August 2018

I think part (iv) of the fundamental theorem of galois theory should say that Gal(F/k) is isomorphic to Gal(E/k)/Gal(E/F)
[Corrected, thanks – T.]

> **Anonymous** · 4 September 2018
>
> I still see the wrong version.
> *[Hopefully fixed for good now – T.]*

Trying to understand the Galois correspondence | What's new - Nevin Manimala's Blog · 29 August 2018

[…] by /u/tick_tock_clock [link] […]

Frank Murphy · 29 August 2018

Reblogged this on fmurphyrng.

hmmm · 29 August 2018

Are there nice geometric reformulations of the classical applications of Galois theory (solvability in radicals of the quintic, ruler-and-compass constructions, and the like), in terms of deck coverings?

> **Terence Tao** · 29 August 2018
>
> For trisection, I did something like this in this previous blog post. For unsolvability of the quintic by radicals, there is a Riemann surface argument of Arnold which one can think of as a covering space argument.

francisrlb · 29 August 2018

In the last paragraph, is there an issue in saying that M_{F} is a quotient of M_{E}, since the point 1 : F -> F, would need to come from a map E -> F?

> **Terence Tao** · 29 August 2018
>
> Oops, you're right, the argument here is a little trickier than in the covering space case (and also I forgot to write up the converse implication for this step). I've rewritten the argument to fix this.

Akshaya · 30 August 2018

Dear Terry,
Could you please try to similarly explain Etale topology, cohomology, point-counting trace formula?

Quick Links | Not Even Wrong · 31 August 2018

[…] Terry Tao has come up with his own take on arithmetic geometry, available here. […]

Anonymous · 3 September 2018

Dear Terry,
I would like to confide with you some words:Last night, I had a strange dream.I found that you

solved 6 clay millenium problems that I could arrange the order of them below:
*1)Yang-Mills existence and mass gaps
*2)Navier-Stokes existence and smoothness
*3)Riemann hypothesis
*4)Birch-Swinerton Dyer conjecture
*5)Hodge conjecture
*6)P vs NP
In all of them, I found that Yang Mills you solved very quickly.You had too many proofs for Navier Stokes.Riemann is the best difficult problem in all of 7 ,because I found that you spend a lot of time doing it.
Aha,there are many things that science never can prove.Many great inventions which comes from a dream.I do not need anyone believe this.Pro.Tao will see my dream is right

 **Nick** · 4 September 2018

Terry: Great exposition. However, in Example 2 (under Theorem 1), I don't think any intermediate field of a cyclotomic extension is another cyclotomic extension. For example, if you take the fixed field of a cyclotomic extension under complex conjugation, it will consist of real numbers and so generally is not a cyclotomic extension.
You probably meant to say "some intermediate fields are of the form".
[Oops, had overlooked that point. Corrected now, thanks – T.]

 **mathcination** · 7 September 2018

I assume this is strongly related to the following book:
Khovanskii: Galois Theory, Coverings, and Riemann Surfaces
https://www.springer.com/de/book/9783642388408

 **Terence Tao** · 7 September 2018

Yes, I should have mentioned some references on this (I am partial to Szamuely's "Galois groups and fundamental groups", for instance). For fields of meromorphic functions on Riemann surfaces and their finite covers there is a very close correspondence between the Galois groups of the fields and the (profinite completion of the) fundamental groups of the Riemann surfaces, as is discussed in Szamuely's book (and presumably also in Khovanskii, which I have not read). For more general schemes than this, the philosophy in most of the literature is to try to move away from the topological, path-based perspective and try to unify these topics in a more abstract setting (e.g. using Grothendieck's functorial point of view). My aim here was slightly different, namely to try to recast Galois theory using the path-based framework. Probably this framework is nowhere near as general as the more category-theoretic approaches, but I found it to be helpful in allowing one to transfer topological intuition to more algebraic settings (as opposed to vice versa).

 **Samuel Vidal** · 9 September 2018

Hi, Sorry if I'm wrong, but I think there is a little typo in example 2. The nth cyclotomic field is an extension of degree $\phi(n)$ over $\mathbb{Q}$.
For example $\mathbb{Q}(\mu_4) = \mathbb{Q}(i)$ is of degree 2. I really like the clarity of your blog. Regards
[Corrected, thanks – T.]

 **Aidan Backus** · 18 October 2020

I think there's a typo in Theorem 4, iv; namely that $Gal(N/L) \cong Gal(M/L)/Gal(M/N)$ rather than $Gal(M/L)/Gal(N/L)$.

[Corrected, thanks – T.]